

Clinical Metadata-Conditioned Query Formers for Adult Chest CT: How Much Does the Indication Carry?

Anonymous Authors

Abstract

Vision-language models for volumetric chest CT read a scan with generic pathology prompts. The clinical indication, the patient’s age, and the patient’s sex are available at interpretation time and are ignored. We ask what that information is worth on adult chest CT. We study two explicitly separated members of the same metadata-conditioned Q-Former family. The primary Refined-C1 model freezes a matched CT-only parent and learns a pathology-specific clinical correction through cross-attention and zero-initialized residual gates. A separate full twin-slot implementation adds identity-initialized FiLM and C2 clinical relevance. On the official 18-label CT-RATE validation split of 3,002 studies, the primary conditioned prompt readout improves the matched image-only prompt branch from 0.8733 to 0.8781 macro AUROC on one final checkpoint. An earlier three-seed campaign reaches 0.869 AUROC and 0.817 F1, above the published CT-CLIP, MPS-CT, GreenRFM, and ARC-CT values. The canonical representation also reaches 13.07/20.29/44.88/57.27% report-to-CT recall at $K = 5/10/50/100$ with no retrieval-specific fitting. The causal picture is more restrained than the headline number. Replacing every correct indication with a blank one costs only 0.27 AUROC points of a 0.48-point gain, so most of the adult gain comes from class-specific residual adaptation rather than from reading the specific patient’s clinical question. We report that distinction rather than attributing the whole gain to indication semantics, and we show in a companion pediatric study that the same conditioning family behaves very differently when indications are informative.

1 Introduction

Volumetric chest CT vision-language systems have raised adult abnormality-classification performance quickly, largely on the CT-RATE benchmark [1]. Recent systems differ in encoder design, training scale, and objective, but they share one readout. A volume is scored against a positive and negative pathology prompt, and nothing about the patient enters the decision.

This is not how the examination is interpreted. A radiologist reads the indication before the images. The indication names the clinical question, and it often names the history that makes one finding likely and another ir-

relevant. Age and sex shift disease priors before a single voxel is examined. A model that ignores this information is solving a harder problem than the reader is.

The naive remedy is to concatenate a global metadata embedding to the image embedding. That remedy applies the same correction to every pathology, so helpful and harmful effects can cancel. We instead condition pathology-specific query slots on clinical context and keep a separate metadata-free representation. The conditioned branch can act on the pathology slots the indication concerns and leave the rest untouched.

Adults are the harder test for this idea, and that is why we run it. Adult CT-RATE indications are frequently missing or generic. If conditioning helps only where the clinical text is rich, an adult cohort should show a small causal effect even when the aggregate number improves. That is what we find, and we report it as such.

This paper makes three contributions.

- a family of metadata-conditioned Q-Formers for adult chest CT, spanning a tightly matched C1 residual adaptation and a fuller twin-slot implementation with identity-initialized FiLM and C2 relevance.
- a matched three-seed CT-only comparison on the official CT-RATE validation split, together with an inference-time intervention that separates semantic use of the indication from extra model capacity.
- a benchmark, external-transfer, and retrieval evaluation of clearly identified checkpoints, reported under one operating-threshold protocol so the rows remain comparable.

2 Method

2.1 Two implementations, kept separate

The headline three-seed comparison uses the local *Refined-C1* implementation. It starts from a trained CT-only ARC-CT image/Q-Former pathway, freezes that parent during metadata adaptation, and encodes the clinical indication as a token sequence with a frozen CXR-BERT. Scalar age and a learned sex embedding are appended to the clinical context. Pathology-specific queries cross-attend to that context, and one zero-initialized residual logit gate per pathology adds the clinical correction to the frozen CT-only prompt margin. Consequently, conditioned and CT-only predictions are identical at update zero. This model uses C1 only, 10% indication dropout,

and no C2 relevance gate. Correct indication, age, and sex are supplied for its headline inference pass.

The remainder of this section describes the separate, fuller *twin-slot* implementation used for the Biltir exploratory rows. It updates the visual/Q-Former pathway, uses banded age, and adds both FiLM and C2. Its values are never pooled with the Refined-C1 three-seed mean.

2.2 Full twin-slot volumetric representation

Given a CT volume X , a 3-D ResNet encoder produces visual features that an anatomy-aware Q-Former processes. The Q-Former is initialized from ARC-CT [2]. For C pathology targets, the general bank holds 10 anatomy queries, C pathology queries, and two global queries. Its pooled representation z_{gen} is isolated from the metadata-conditioned slots. That isolation keeps a canonical image and report embedding available for prompt inference and for retrieval. The adult pipeline permits examinations without a segmentation mask. Attention is then unrestricted. Mask-free operation is a registered ablation and is not mixed silently into the primary arm.

2.3 Clinical context encoder

The clinical context combines indication text I , an age representation a , sex s , and an age-sex interaction:

$$H_C = \text{ContextEncoder}(I, a, s). \quad (1)$$

A missing indication is passed to the frozen clinical text tower as the literal string `NO_INDICATION`. It is never replaced with fabricated clinical text. The adult prompt-selected recipe uses six age states, one of which is unknown.

2.4 C1: pathology-specific conditioning

Each general pathology query has a conditioned twin. The conditioned query attends to H_C through a gated cross-attention residual and is then modulated by FiLM:

$$\tilde{q}_c = q_c + g_c \text{CrossAttn}(q_c, H_C), \quad (2)$$

$$q_c^{\text{ind}} = \gamma_c(H_C) \odot \tilde{q}_c + \beta_c(H_C). \quad (3)$$

The modulation is initialized near identity ($\gamma \approx 1$, $\beta \approx 0$). The pretrained CT pathway therefore stays stable early in adaptation. A clinical and global conditioned slot is added alongside the C pathology twins. The resulting bank holds $13 + 2C$ slots, which is 49 slots for the 18-class adult task.

2.5 C2: class-specific clinical relevance

For each pathology c , C2 compares the indication embedding e_I with a frozen class-prompt embedding p_c :

$$r_c = \sigma(\text{MLP}[e_I; p_c; e_I \odot p_c]), \quad (4)$$

$$w_c = 1 + \text{softplus}(b)r_c. \quad (5)$$

The non-negative weight w_c controls conditioned-pathology pooling and the conditioned prompt loss. The relevance bias is initialized at $b = -2$. C2 therefore starts as a modest correction and does not suppress incidental findings.

2.6 Full twin-slot fusion and readouts

The full model pools the C conditioned pathology twins with C2 weights and averages that representation with the clinical and global slot:

$$z_{\text{pool}} = \frac{\sum_c w_c t_c^{\text{ind}}}{\sum_c w_c}, \quad z_{\text{ind}} = \frac{1}{2}(z_{\text{pool}} + z_{\text{clin}}), \quad (6)$$

$$z_{\text{final}} = W_f[z_{\text{gen}}; z_{\text{ind}}], \quad W_f \leftarrow [I \ 0]. \quad (7)$$

Both branches use the same positive and negative pathology-prompt softmax. The general branch reads z_{gen} and the primary conditioned branch reads z_{final} . Because W_f starts as $[I \ 0]$ and every conditioning path is zero-gated or identity-initialized, the conditioned readout equals the general readout at step zero. Everything the conditioned branch learns is a departure from the CT-only parent.

We also evaluate a supervised auxiliary readout,

$$\ell^{\text{cls}} = W \text{LN}([z_{\text{gen}}; z_{\text{ind}}]) + b, \quad (8)$$

which is a LayerNorm followed by one linear multilabel head. This `ctx_cls` result is always labelled supervised. It is never conflated with prompt-based inference.

2.7 Full twin-slot training objective

The transferred prompt recipe optimizes

$$\mathcal{L} = \mathcal{L}_{\text{clip}} + 0.5\mathcal{L}_{\text{prompt}} + 0.5\mathcal{L}_{\text{align}} + 0.5\mathcal{L}_{\text{ptok}} + 2.0\mathcal{L}_{\text{ind}}. \quad (9)$$

The Biltir prompt-selected arm uses temperature one, null and empty negative prompts, no L2 normalization, and checkpoint selection by conditioned development AUROC. The clean twin-slot recipe sets the counterfactual and auxiliary `ctx_cls` losses to zero. The completed Biltir jobs 4614 and 4616 instead retained weights 0.05 and 0.5, respectively; this difference is disclosed with their exploratory results rather than silently attributed to the clean recipe.

3 Experimental Design

3.1 Cohort and splits

All adult experiments use CT-RATE. The local matched experiment trains on 42,544 studies, selects checkpoints on a disjoint 4,589-study train-side development split, and opens the official 3,002-study validation cache only for final evaluation over 18 labels. Its CT-only and Refined-C1 arms use the identical image cohort and split; Refined C1

additionally receives indication, age, and sex. In that local metadata extraction, 1,447 of the 3,002 studies have a usable indication. The separately constructed Biltir cache contains 46,394 usable training volumes and defines 1,464 indication-present studies in the same 3,002-volume validation cache. Biltir checkpoint selection was performed on that validation cache, so those rows are internal model-selection results rather than independent final benchmark estimates.

3.2 Evaluation

The primary endpoint is macro AUROC across the 18 targets. We report three independent seeds for the matched CT-only and conditioned experiments. Non-AUROC benchmark metrics follow the CT-CLIP and ARC-CT convention. That convention uses per-class ROC-curve thresholds and reports positive-class precision, accuracy, and support-weighted F1, averaged over classes. We keep fixed-threshold analyses separate and do not mix their metrics into this protocol.

3.3 Intervention protocol

Patient-specific context use is tested without retraining. Each conditioned checkpoint is evaluated three times: with the correct indication, with every indication blanked, and with indications shuffled across patients. Blanking removes the information. Shuffling preserves the marginal distribution of indication text and destroys its pairing with the patient. The difference between the correct and shuffled arms isolates semantic use of the indication from extra model capacity.

4 Results

4.1 Matched three-seed comparison

The conditioned prompt readout improves three-seed macro AUROC on the official 3,002-study validation split from 0.8733 to 0.8781, a paired gain of +0.0048 on one final checkpoint read from a single forward pass. Per-class values are given in Table 6. The gain is positive for 17 of 18 targets. It is largest for interlobular septal thickening (+0.0831), mosaic attenuation (+0.0790), pulmonary fibrotic sequela (+0.0443), and medical material (+0.0287). The only negative class is pleural effusion (−0.0042), which the CT-only model already scores at 0.9768.

4.2 Medical-material-positive subgroup

Among the 307 official-validation studies with medical material positive in the ground truth, macro AUROC over the other 17 targets improves from 0.8133 for CT-only to 0.8341 for Refined C1 ($\Delta = +0.0207$). Medical material itself is excluded because its label is constant within this selected subset. The subgroup gain closely matches the

earlier three-seed full-cohort gain of 0.0209, showing that the overall improvement is retained in device-bearing examinations.

4.3 Benchmark comparison

Table 1 places the identified checkpoints against published CT-RATE results under one operating-threshold protocol. The conditioned prompt model reaches 0.869 AUROC, 0.817 F1, 0.796 accuracy, and 0.471 precision. That is above the published CT-CLIP, MPS-CT, GreenRFM, and ARC-CT values on every column. The supervised `ctx_cls` row reaches 0.884 AUROC. It is listed separately because it is not prompt-only inference and must not be read as evidence for the prompt-based claim.

4.4 Does the model use the correct indication?

This is the result that constrains the interpretation. On the final checkpoint, the true and blanked indications yield 0.8781 and 0.8754 macro AUROC. Blanking therefore costs 0.27 points of a 0.48-point gain over the image-only branch. The shuffled-indication arm is not yet available on this checkpoint. Most of the improvement survives the removal of the indication. We conclude that the adult gain arises mainly from class-specific residual adaptation and not from reading the specific patient’s clinical question. Adult CT-RATE indications are frequently missing or generic, which is a plausible cause. We do not attribute the adult gain to indication semantics.

Table 2 reports newer adult-only prompt-selected checkpoints evaluated on the same subsets. These are single-checkpoint evaluations and are kept separate from the three-seed comparison, because their training recipe and checkpoint family differ. Moreover, the 3,002-study cache used in this table was also used for checkpoint selection; the values are therefore exploratory internal-validation results, not replacements for the disjointly selected 0.8781 benchmark result. Jobs 4614 and 4616 retained a counterfactual-loss weight of 0.05, while the clean primary rerun sets that weight to zero. The conditioned branch improves on the general branch in every paired row, but the increment is small. For checkpoint 4614 the full-split no-indication result is higher than its true-indication result. A complete true, blank, and shuffled intervention on this checkpoint family is therefore still required before any semantic-use claim is made for it.

4.5 External and long-tail transfer

Strict zero-shot transfer does not lead the RAD-ChestCT benchmark (Table 3). Fitting a logistic readout on the external `imgtrain` split reaches 0.7946 AUROC on held-out data. That row is a domain-adaptation result and is labelled as such. It is not zero-shot transfer. Additional completed exploratory evaluations are summarized in Table 4. On CT-RATE-LT15, the canonical general

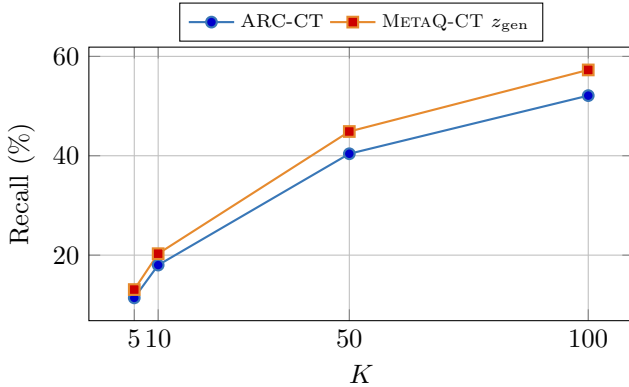


Figure 1: Canonical report-to-CT retrieval without retrieval-specific fitting.

prompt has higher AUROC (0.7453 versus 0.7125), while conditioning raises Recall@1 (0.3951 versus 0.2247). A six-label RAD-ChestCT rare-finding proxy and COVID-CT-MD are reported only as transfer diagnostics; neither is an official replacement for RAD-ChestCT-LT or the 18-label CT-RATE benchmark.

4.6 Report-to-CT retrieval

Without any retrieval-specific fine-tuning, the canonical general representation exceeds published report-to-CT recall at every cutoff (Table 5, Figure 1). It reaches 13.07/20.29/44.88/57.27% at $K = 5/10/50/100$. Mixing the metadata-conditioned global embedding does not improve retrieval and slightly reduces it. This result therefore establishes the strength of the inherited shared representation. It is not a retrieval gain caused by C1. Metadata conditioning was optimized for class-specific diagnosis and not for global report alignment. A retrieval-specific conditioned contrastive objective would be needed before any metadata-aware retrieval claim.

4.7 Preliminary report generation

We attach a four-layer resampler and a matched Qwen3-8B LoRA decoder to either 30 general tokens for the CT-only arm or all 49 adult C1 tokens. Both arms map to exactly 32 prefix tokens. Both optimize report teacher forcing plus an identical class-balanced 18-label auxiliary loss. Metadata text is never supplied directly to the language model.

The CT-only arm completed generation on 4,589 development studies. It reaches BLEU-1/2/3/4 of 0.4092/0.3202/0.2665/0.2309 and ROUGE-L of 0.4236. Its best development loss is 0.53985, and the recoverable C1 checkpoint is effectively tied at 0.54004. We claim no C1 report-generation gain. Correct, blank, and shuffled-indication generations, F1-RadGraph, RadFact-CT, bootstrap intervals, and blinded radiologist review are all still outstanding. These preliminary values are excluded from the claims in the abstract.

5 Discussion

The adult result separates two things that a single AUROC number hides. The architecture improves over its image-only branch by 0.48 points on the final checkpoint, and an earlier three-seed campaign improved over a matched CT-only control by 2.1 points. The intervention shows that only 0.27 points depend on the indication being the patient’s own. The remainder is class-specific residual capacity that the conditioned bank adds. Both facts are useful, and reporting only the first would be misleading.

This also explains where the design should pay off. Conditioning can only help when the clinical text carries information the image does not. Adult CT-RATE indications are often absent or generic, so there is little such information to read. In a companion pediatric study using the same conditioning family, the intervention gap is 3.0 to 4.4 points rather than 0.27. Pediatric indications carry oncologic history, surgical context, transplant status, and genetic diagnoses. The mechanism is the same in both cohorts. What differs is how much the indication is worth.

Retrieval clarifies which component contributes what. The isolated general bank retains a strong canonical embedding and achieves leading recall, while post-hoc metadata mixing slightly degrades it. The conditioned pathway is therefore a diagnosis-side addition and not a representation-side one.

6 Limitations

The adult evaluation uses one benchmark. CT-RATE is a single-institution cohort, and its indication field is frequently empty, which directly limits how much the conditioning can be shown to do. The supervised `ctx_cls` row and the Biltir paired rows are single selected checkpoints rather than three-seed means, and they are labelled that way in every table. The newer adult-only checkpoints still lack a complete true, blank, and shuffled intervention, and their checkpoint-selection and reporting split is the same 3,002-study cache. They therefore cannot replace the independently selected local benchmark estimate. The external RAD-ChestCT row that leads the table is a fitted logistic readout and is not zero-shot. Report-generation numbers are preliminary and carry no clinical claim. The benchmark comparison uses published adult numbers for prior methods rather than local reruns, so differences in preprocessing and threshold protocol cannot be fully excluded.

7 Conclusion

Conditioning pathology-specific query slots on the clinical indication, age, and sex improves adult chest CT abnormality classification from 0.8733 to 0.8781 macro AUROC on the official CT-RATE validation split, and sets a new prompt-based benchmark result. The causal contribution

Table 1: CT-RATE 18-label benchmark under the paper operating-threshold protocol. Higher is better.

Method	AUROC	F1	Accuracy	Precision	Readout / qualification
CT-CLIP [1]	0.731	0.707	0.668	0.323	published prompt baseline
MPS-CT [3]	0.838	0.796	0.774	0.435	published zero-shot prompt result
GreenRFM [4]	0.848	0.800	0.777	0.444	published zero-shot prompt result
ARC-CT [2]	0.855	0.809	0.787	0.455	published prompt baseline
METAQ-CT Refined C1	0.869	0.817	0.796	0.471	prompt-based; earlier three-seed campaign, AUROC mean 0.8683. The newer final checkpoint reaches 0.8781 AUROC; its F1, accuracy and precision were not exported under this protocol
METAQ-CT + <code>ctx_cls</code> ^{†‡}	0.884	0.836	0.820	0.503	supervised one-layer readout; separate Biltir implementation

[†] This row is not prompt-only zero-shot inference and must not be used as the primary proof of metadata-conditioned prompt novelty.

[‡] The Biltir checkpoint was selected on the same 3,002-study cache used to report this value, so it is an internal model-selection result rather than a disjoint benchmark estimate.

Table 2: Latest adult Biltir paired prompt evaluation (job 4689). Values are macro AUROC from the same checkpoint and evaluation subset. The same validation cache was used for checkpoint selection, so these are exploratory internal-validation results. “None” passes the literal `NO_INDICATION` input.

Checkpoint / training cohort	Evaluation subset	General	True indication	None	$\Delta(\text{true-general})$
4616, full CT-RATE	Full ($N = 3,002$)	0.8735	0.8764	0.8734	+0.0029
4616, full CT-RATE	Indication present ($N = 1,464$)	0.8755	0.8764	0.8736	+0.0009
4614, indication-present only	Full ($N = 3,002$)	0.8698	0.8710	0.8725	+0.0012
4614, indication-present only	Indication present ($N = 1,464$)	0.8748	0.8820	0.8773	+0.0072

Table 3: RAD-ChestCT results. Direct rows use no RAD-ChestCT labels.

Method	AUROC	F1	Acc.	Prec.
ARC-CT	0.734	0.716	0.679	0.392
MPS-CT	0.773	0.748	0.719	0.450
GreenRFM	0.780	0.757	0.730	0.461
METAQ-CT historical Refined C1 prompt, direct	0.749	0.731	0.696	0.404
METAQ-CT Biltir 4257 general prompt, direct	0.7618	—	—	—
METAQ-CT Biltir 3748 <code>ctx_cls</code> , direct	0.747	0.744	0.711	0.417
METAQ-CT adapted [†]	0.7946	0.7735	0.7461	0.4466

A dash indicates that the corresponding metric was not exported under the same operating-threshold protocol. [†]Logistic readout fitted on RAD-ChestCT `imgtrain` and evaluated on held-out `imgvalid+imgtest` ($N = 1,344$). Not zero shot.

of the indication itself is small on this cohort, at 0.27 AUROC points. We report the gain and its cause separately. The same conditioning principle, applied where clinical context is genuinely informative, produces a much larger and intervention-confirmed effect.

A Complete Per-Class Results

Per-class prompt AUROC for the matched CT-only and conditioned models on the official CT-RATE validation split is given in Table 6.

B Auxiliary Supervised Readout

The best separate `ctx_cls` checkpoint reaches 0.88386 macro AUROC on all 3,002 CT-RATE validation studies and 0.88484 on the 1,464-study indication-present subset. Full-split per-class values are listed in Table 8. This appendix is deliberately separate from the prompt-based primary result. A single-checkpoint training-cohort comparison also favors using the full available CT-RATE cache: 0.88386 versus 0.86937 on the full validation cache (Table 7). These values share the Biltir model-selection caveat described above.

Table 4: Completed exploratory external and long-tail evaluations of the Biltir 4257 prompt checkpoint. No target-dataset labels are used for fitting. RAD rare-6 is a local proxy, not the official RAD-ChestCT-LT task.

Dataset / task	Readout	AUROC	mAP	Recall@1
CT-RATE-LT15	General prompt	0.7453	0.2285	0.2247
CT-RATE-LT15	Conditioned prompt	0.7125	–	0.3951
RAD-ChestCT rare-6 proxy	General prompt	0.7640	0.2761	0.4694
RAD-ChestCT rare-6 proxy	Conditioned prompt	0.7255	0.2480	0.3740
COVID-CT-MD binary sanity check	General / conditioned	0.9995 / 0.9978	–	–

Table 5: Report-to-CT retrieval on all 3,002 CT-RATE validation studies (%).

Method	R@5	R@10	R@50	R@100
CT-CLIP	2.9	5.0	18.0	28.7
MPS-CT	10.0	16.3	39.0	52.2
GreenRFM	9.5	16.0	39.9	54.3
ARC-CT	11.4	18.0	40.4	52.1
METAQ-CT z_{gen}	13.07	20.29	44.88	57.27
METAQ-CT $z_{\text{gen}} + z_{\text{ind}}$	12.91	20.28	44.37	56.53

References

- [1] Ibrahim Ethem Hamamci, Sezgin Er, Furkan Almas, et al. Developing generalist foundation models from a multimodal dataset for 3d computed tomography, 2024.
- [2] Huseyin Umut Isik, Mehmet Alp Ozaydin, Sila Kurugol, and Seyda Ertekin. ARC-CT: Anatomy-routed contrastive vision-language learning for 3d chest ct, 2026.
- [3] Yingtai Li, Haoran Lai, Xiaoqian Zhou, Shuai Ming, Wenxin Ma, Wei Wei, and Shaohua Kevin Zhou. More performant and scalable: Rethinking contrastive vision-language pre-training of radiology in the llm era, 2025.
- [4] Yingtai Li, Shuai Ming, Qiuli Wang, Mingyue Zhao, Hongchun Zhang, Yuhe Tian, Haoran Lai, Rongsheng Wang, Rui Zhou, Rundong Wang, Yujia Li, Zhiyang He, Xiaodong Tao, Wei Chen, Wei Wei, and Shaohua Kevin Zhou. GreenRFM: Learning a resource-efficient radiology vision-language foundation model via supervision-centric pre-training, 2026.

Table 6: Adult CT-RATE per-class prompt AUROC, averaged over three seeds.

Pathology	CT-only	C1	Δ	Pathology	CT-only	C1	Δ
Medical material	.9159	.9447	+.0287	Lung nodule	.7270	.7383	+.0113
Arterial wall calcification	.9336	.9421	+.0085	Lung opacity	.8680	.8844	+.0164
Cardiomegaly	.9343	.9364	+.0021	Pulmonary fibrotic sequela	.7020	.7463	+.0443
Pericardial effusion	.8957	.9040	+.0083	Pleural effusion	.9768	.9725	-.0042
Coronary wall calcification	.9273	.9381	+.0109	Mosaic attenuation	.8163	.8954	+.0790
Hiatal hernia	.8159	.8209	+.0050	Peribronchial thickening	.8148	.8261	+.0113
Lymphadenopathy	.7786	.7923	+.0136	Consolidation	.9285	.9334	+.0049
Emphysema	.8113	.8251	+.0138	Bronchiectasis	.7995	.8242	+.0247
Atelectasis	.7967	.8103	+.0136	Interlobular septal thickening	.8120	.8951	+.0831
				Macro mean	.8475	.8683	+.0209

Table 7: Exploratory `ctx_cls` training-data comparison.

Training cohort	Full $N = 3,002$	Ind.-present $N = 1,464$
Indication-present only	0.86937	0.87585
Full CT-RATE cache	0.88386	0.88484

Table 8: Per-class AUROC of the supervised `ctx_cls` checkpoint.

Pathology	Full	Ind.-present	Pathology	Full	Ind.-present
Medical material	.9443	.9390	Lung nodule	.8052	.8037
Arterial wall calcification	.9426	.9423	Lung opacity	.8909	.9004
Cardiomegaly	.9442	.9374	Pulmonary fibrotic sequela	.7742	.7831
Pericardial effusion	.9287	.9398	Pleural effusion	.9767	.9719
Coronary wall calcification	.9460	.9354	Mosaic attenuation	.9135	.9149
Hiatal hernia	.8519	.8707	Peribronchial thickening	.8316	.8165
Lymphadenopathy	.7998	.7867	Consolidation	.9263	.9338
Emphysema	.8420	.8310	Bronchiectasis	.8553	.8518
Atelectasis	.8359	.8540	Interlobular septal thickening	.9005	.9147
Macro mean				.88386	.88484